Perceptual recoverability, articulatory efficiency & speech flexibility

- Speech flexibility (i.e., motor equivalence; cf. Perrier & Fuchs 2015) suggests that the same target can be achieved through an array of motor movements.
- Certain degrees of freedom exist, allowing for an intended speech target to be achieved via an array of articulatory synergies.
    - Perceptual recoverability and articulatory efficiency could be acheived at the same time.
- Consequently, such flexibility may be a window for the speaker to react differently in the face of variable information-theoretic constraints in the articulatory & acoustic domains.

Perrier, P., and Fuchs, S. (2015) Motor Equivalence in Speech Production. In M. A. Redford (eds.), *The Handbook of Speech Production*, (225–247). New Jersey: John Wiley & Sons.

Research questions

1. Does a target phone's distinctiveness as a function of the preceding context phone positively correlate with the bigram's surprisal, given its context within a word?

2. If so, would flexibility allow for an enhanced positive surprisal effect in the acoustic domain while leading to a reduced positive effect/a negative effect of surprisal in the articulatory domain?